

Raj Khatri

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PROFESSIONAL SUMMARY

Final-year Industrial & Production Engineering student working in data analysis and manufacturing analytics. At Tata Motors, traced a live Body-in-White production defect to a supplier-side dimensional cause using regression-based trend analysis, confirmed it by direct measurement on the shop floor, and designed a containment procedure that needed no line stoppage. Currently a part-time backend developer building the PostgreSQL data layer for a live K–12 ed-tech platform. Strongest in Python, SQL, and applied statistics.

EDUCATION

B.Tech, Industrial & Production Engineering
College of Technology, G.B. Pant University of Agriculture & Technology, Pantnagar

Uttarakhand, India
August 2023 – August 2027

Senior Secondary (XII), CBSE · Maheshwari Public School, Kota · 75.2%
Secondary (X), CBSE · Peace Public School, Gopeshwar · 97.2%

2022
2020

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL / MySQL), JavaScript, Bash

Data & Analytics: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter, Excel (Advanced / VBA), Power BI

Backend & Tools: Flask, SQLAlchemy, Supabase, Next.js, Streamlit, Git/GitHub, Vercel, VS Code

Methods: Statistical modeling & hypothesis testing, root-cause analysis, DFMEA, Six Sigma, exploratory data analysis, demand forecasting, dashboard development

EXPERIENCE

Tata Motors
Industrial Trainee — Data Analytics
Project: "Predictive Quality Intelligence — A Data-Driven Framework for BIW Root-Cause Diagnosis"

Pantnagar, Uttarakhand (**On-site**)
03 July 2026 – 03 August 2026

- Identified a supplier-side dimensional defect as the root cause of a recurring Body-in-White quality issue, ruling out on-line equipment wear.
- Reached the finding by applying linear regression with R^2 confidence scoring across ~300 inspection data points, then validating it through direct physical measurement on the shop floor.
- Designed a containment procedure deployable without production stoppage or vendor dependency.
- Built an offline, interactive HTML/JS diagnostic dashboard tracking defect trends across 10 monitored quality metrics.
- Authored a 20+ section technical report benchmarking the findings against DFMEA, Six Sigma, and peer-reviewed manufacturing literature.

Voldebug Innovations Pvt. Ltd.
Backend Developer (Part-time)

Remote
22 August 2026 – Present

- Migrating a live K–12 AI-literacy platform from browser-local persistence to a Supabase (PostgreSQL) backend — schema design, row-level security policies, and role-based authentication for student, teacher, and principal accounts.
- Implementing server-side APIs in Next.js (App Router) for XP/badge progression, class-code enrollment, and teacher analytics, replacing client-only state with a shared data layer.
- Establishing database migrations, staging/production environment separation, and a repeatable deployment workflow on Vercel.

Intern

01 July 2026 – 31 July 2026

- Built the platform's first working version — a gamified AI-literacy SPA spanning student, teacher, and principal roles, with a curriculum-linked prompt library and an XP/badge/streak progression system.
- Developed AI-assisted lesson-planning tools and a principal-facing analytics dashboard; implemented Hindi/English bilingual support and downloadable student certificates.
- Diagnosed and fixed browser back-navigation bugs using hash-based query parameters, and resolved orphaned timer state affecting session tracking.
- Authored the phased Supabase + Next.js migration plan now being implemented in the backend role above.

PROJECTS

Supply Chain Optimization Engine *Python, Scikit-learn, Pandas, NumPy*

- Trained a Random Forest model on historical demand to generate daily forecasts in place of manual estimation.
- Implemented statistical safety-stock calculations to buffer against demand and lead-time volatility.
- Built replenishment logic around Economic Order Quantity (EOQ) sizing with reorder-point triggers.

Automated Fleet Routing Simulation *Python, Google OR-Tools, Folium/Leaflet*

- Modeled multi-stop delivery routing as a combinatorial optimization problem, applying heuristics to reach near-optimal stop sequences.
- Computed distance matrices from geospatial coordinates and fed them into a Constraint Programming solver to maximize fleet utilization.
- Added a workload-balancing objective to distribute stops evenly across drivers; rendered resulting routes on interactive Leaflet/Folium maps.

Nasdaq Tech Stocks EDA *Python, Pandas, yfinance, Matplotlib, Seaborn*

- Analyzed Apple, Microsoft, Amazon, and Alphabet price history (2020–2025) for trend, volatility, and drawdown behavior.
- Computed pairwise correlations of 0.58–0.68 across the four stocks, quantifying their co-movement during the period.

CERTIFICATIONS

Forage Job Simulations: J.P. Morgan (Quantitative Research) · Deloitte Australia (Data Analytics) · Mastercard (Cybersecurity)

2026